Physics 101, Problem Sheet 4: Forces on a Slope

Attempt all questions and show every step of your working. Hand in your solutions at the Tuesday tutorial.

Formula sheet

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v = u + a*t
s = u*t + 0.5*a*t**2
v**2 = u**2 + 2*a*s
F = m*a
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Questions

1. A 2.0 kg block slides from rest down a smooth slope inclined at 30° to the horizontal.
   (a) Draw a free-body diagram for the block.
   (b) Show that its acceleration down the slope is 4.9 m/s².
   (c) Find its speed after it has travelled 3.0 m along the slope.

2. A trolley is released on a slope and timed over a fixed distance of 1.20 m at several angles θ.

Trial	Angle (°)	Time (s)
1	10	1.18
2	15	0.97
3	20	0.84
4	25	0.75
5	30	0.69

   (a) Plot a graph of $1/t^2$ against $\sin\theta$.
   (b) Use the gradient to estimate the acceleration due to gravity.

3. A car of mass 1200 kg is parked on a hill. The coefficient of static friction between the tyres and the road is $\mu = 0.40$.
   (a) Calculate the largest slope angle at which the car stays at rest. Your answer should explain which of the following is true:
      (i) the friction force equals the component of weight down the slope;
      (ii) the friction force is at its limiting value μN;
      (iii) the normal reaction equals the weight of the car.
   (b) State one assumption that makes your answer an upper limit.

4. A skier of mass 70 kg descends a 200 m slope that falls through 40 m.
   (a) Find the gain in kinetic energy if friction is ignored.
   (b) Her actual speed at the bottom is 18 m/s. How much energy was lost?

Notes on marking

Marks are given for correct method as well as for correct numerical answers. Answers to selected questions will be posted on the course page on Friday.